Solar and Gas (SNG)

Group :3

Group name : Team Helios

Product name: SNG

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Course: Engineering Design CPE4813

Professor Johnathan Voition

Table of Contents

Contents

1. Executive Summary
2. Introduction
3. Needs Being Addressed
4. Literature and Patent Information
5. Market Analysis and Market Strategy
6. Engineering Design Constraints
7. Product Requirements/Constraints
8. Engineering Codes and Standard
9. Design Concepts
10. Major (critical) Components
11. Hardware Block Diagram
12. Enclosure Design
13. Software
14. Major Problems & Conclusions

Section 1 EXECUTIVE SUMMARY

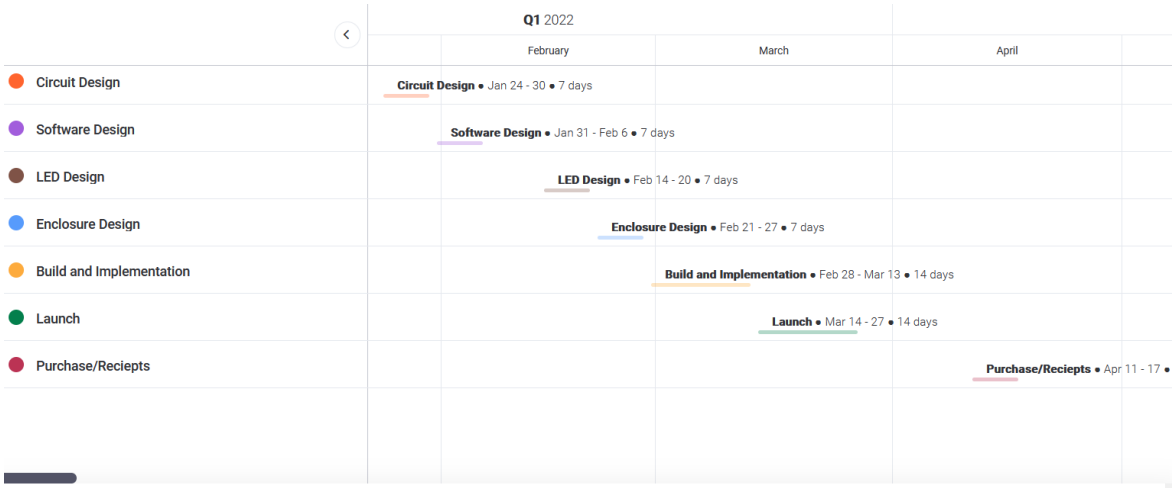
The solar-powered light fixture w/ gas sensor design is a product of Kinghe Systems. The following is a brief description of its overall design, how it will be implemented, its core uses, and its conception in the market.

This solar powered light fixture was designed intuitively in Design 1 by Team 3 (Helios). This design was initially designed to alert the customer to harmful gasses emitted in the air through various means, such as: vehicle exhaust, fires, and mixtures of other gasses that can be harmful, via solar panels. The overall design has a raspberry pi 3b+, Arduino 2560, MQ2 gas sensor, MQ9 gas sensor, a PIR sensor, light sensor, potentiometer, solar panels, 12V battery, and a solar charge controller. The solar panels receive sunlight which will go to the solar charger, which is powered by a 12V battery; this, in turn, will give power to the raspberry pi and the Arduino 2560 microcontrollers. These microcontrollers will send power to the gas sensors so they can detect harmful gasses by turning red upon gas detection and a steady green light when no gas is detected. When semester 2 of Engineering design started, team 3 developed this project by re-evaluating the original design, design constraints, and budget. The Arduino 2560 will act as the main microcontroller for this project, due to Python computer language. The customer will have the option of home installation or along major routes of travel. This device is user friendly, cost friendly, and very environmentally friendly and successful. What we proposed at the end of last semester was to have our project working with just the raspberry pi, but eventually we went with both the raspberry pi and the Arduino 2560, due to coding issues.

Section 2: Introduction

The main focus of the solar powered light fixture with gas sensors is to give the customer an overall view of how solar cells work and its cost-efficiency in everyday life. Air is the most vital source of human function and without it, there will be no life. So, when air is distorted with gas and airborne illnesses such as carbon monoxide, CO2, bird flu, and Covid 19. These gas/illnesses can be fatal if someone has asthma, upper respiratory problems, and lung/nasal issues. Filtering out these airborne gasses is detrimental to everyday life.

Team Schedule



Build and Implementation · Circuit Design · Enclosure Design · LED Design · Launch · Purchase/Reciepts · Research · Software Design

Research	Owner	Subitems	Status	Timeline
This is a subitem				-
Find Solar panel and Battery			Planned	Jan 17 - 23
Raspberry Pi specification: SPI, UART, ADC			Planned	Jan 17 - 23
Find Calibration data to compare			Planned	Jan 17 - 23

Circuit Design	Owner	Subitems	Status	Timeline
Battery Charger			Planned	Jan 24 - 30
Power Converter			Planned	Jan 24 - 30
Solar Panel			Planned	Jan 24 - 30
Voltage Divider			Planned	Jan 24 - 30
Sensor Array			Planned	Jan 24 - 30
Cellular Transciever			Planned	Jan 24 - 30
voltage divider			Planned	Jan 24 - 30

Software Design

	Owner	Subitems	Status	Timeline
Raspberry pi IDE			Planned	Jan 31 - Feb 6
Sensor data read and log			Planned	Jan 31 - Feb 6
Cellular Transciever data			Planned	Jan 31 - Feb 6
Read Calibration			Planned	Jan 31 - Feb 6
Send Alert			Planned	Jan 31 - Feb 6

LED Design

	Owner	Subitems	Status	Timeline
LED's			Planned	Feb 14 - 20
Ligh sensor			Planned	Feb 14 - 20
motion sensor			Planned	Feb 14 - 20
Led driver			Planned	Feb 14 - 20

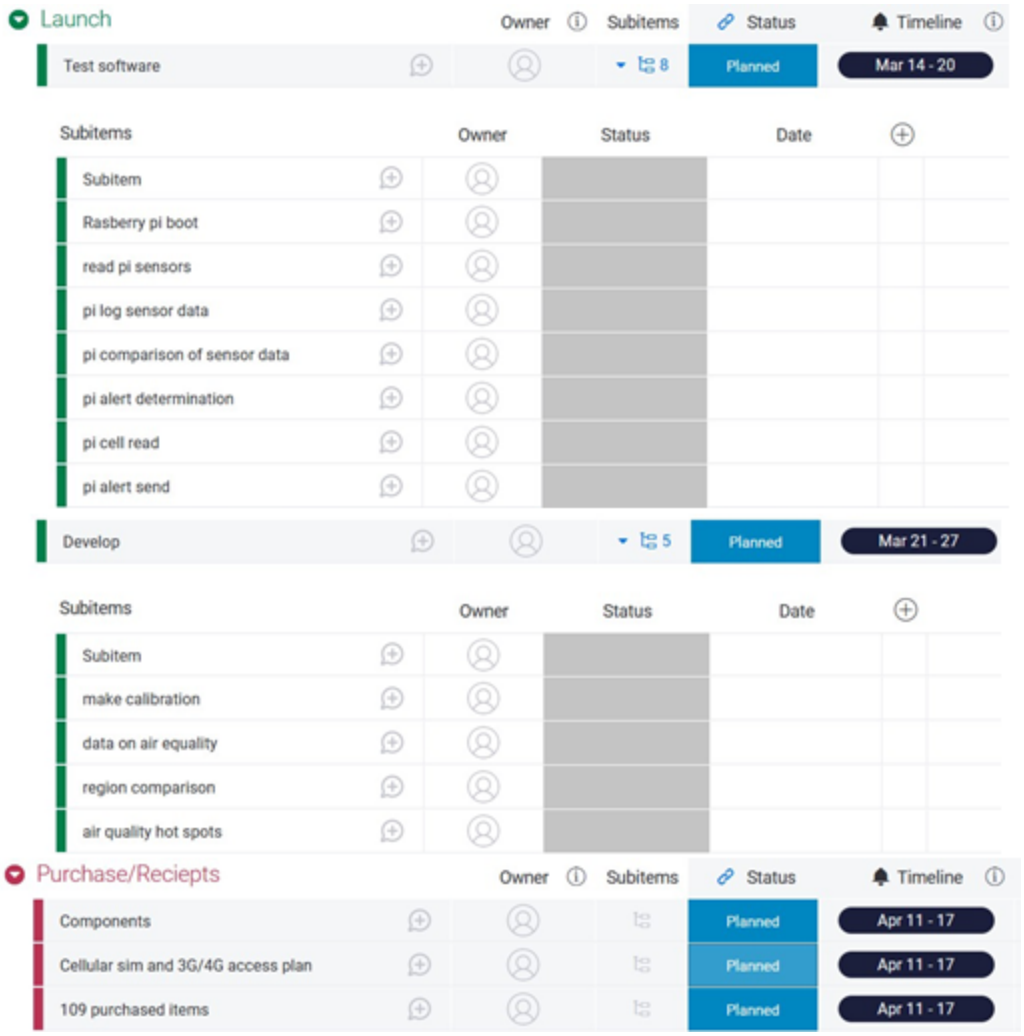
Enclosure Design

	Owner	Subitems	Status	Timeline
Metal or Plastic			Planned	Feb 21 - 27
Component Dimension			Planned	Feb 21 - 27
Component Mounting			Planned	Feb 21 - 27
Venilation			Planned	Feb 21 - 27

Build and Implementation

	Owner	Subitems	Status	Timeline
Build and Implementation		6	Planned	Feb 28 - Mar 6
Build Solar Panel enclosure,			Planned	Feb 28 - Mar 6
Build device enclosure			Planned	Feb 28 - Mar 6
Connect sensors ad cellular transeiver,			Planned	Feb 28 - Mar 6
Build LED enclosure,			Planned	Feb 28 - Mar 6
Connect to Raspberry Pi,			Planned	Feb 28 - Mar 6
Build charger and battery circuit,			Planned	Feb 28 - Mar 6
Test Hardware		7	Planned	Mar 7 - 13

Subitems	Owner	Status	Date
Solar panel voltage			
Battery Charger			
Voltage Divider			
Rasperry Pi power			
Sensors Power			
LED Power			
Test Outside in Sun			



Section 3: Needs Being Addressed

This project's scope is to target the needs of customers who desire better air quality especially during this pandemic. Budgetary constraints are not a big issue due to this project's cost efficiency and ease of use. Computer chips may be an issue for future references because of an international demand on Silicon computer chips.

Section 4: Literature and Patent Search

Since our design has solar powered elements, which are pretty common and modern, our patent search came up with a lot of solar powered light fixture patents, but none with MQ9 and MQ2 gas sensors. A google patent search was done to separate this project from patents already

initiated. As with any new design, patent search is very important to keep an entrepreneur from inventing something that is already patented.

Section 5: Marketing Analysis and Market Strategy

This section will briefly describe the potential of the solar powered light fixture with the gas sensor market and the target audience involved in the modern, tech savvy industry.

The solar powered marketing analysis is very robust and feasible. Our design, as of now, is not ready for manufacture, but the design concept proves promising. Our general audience are mainly those interested in climate control, high air quality, and Ozone layer protection.

Our strategy is to make the prototype user friendly, budget friendly, easy to manufacture, and easy to maintain. We plan to use the present technology available to make our product user friendly to those who may be "technologically challenged" and those who have respiratory issues. The biggest audience for our product are: hospitals, fire departments, police precincts, VIA, and Universities or places of businesses that have motors or engines.

Section 6: Engineering Design Constraints

The engineering design constraints are used to develop and potentially manufacture the solar powered light fixture with gas sensors. These constraints include safety, environmental, manufacturing, and legal constraints. These constraints help make the overall design process a more feasible and user friendly product.

GLOBAL CONSTRAINTS

1. Safety constraints
    - Safety labels printed on the product is always a precautionary measure when production of a newly released product is manufactured.
    - These guidelines should adhere to the IEEE engineering standards.
    - This applies to our project due to the fact that safety is always applied.

2. Legal Constraints
    - All copyrights and trademarks should be reviewed before publishing.
    - This applies to our design especially since there are many on the market that resemble our project.

- Legal constraints do not apply to our project because it adheres to the US ethical standards and engineering codes.

3. Economic constraints
    - The size of the market is big but it will be more expensive compared to the other similar product
    - Getting capital for the product might be difficult due to the competitive nature of the market
    - This constraint does not apply because our product is very inexpensive.

4. Manufacturing constraints
    - Due to shortage of raw materials such as silicon for building chips, it might be more difficult and expensive to get the materials needed for the product.
    - Cost of labor to assembly, transport and inspect product and materials
    - This constraint applies to our project, due to a global shortage of computer and integrated chips.

LOCAL CONSTRAINTS

5. **Schedule constraints**
    - Planning and testing the product given the timeframe we are currently working with might be challenging
    - Delivering the product to thousands of solar panels across the United States on time is challenging due to a shortage of raw materials, so therefore, this applies to our project.

6. **Sustainability constraints**
    - Market sustainability is low due to the product being easy to reverse engineer
    - Possible political consequences as far as regulations on pollution control to fight climate change
    - Therefore, this constraint does not apply.

7. **Life-cycle constraints**
    - The operation of the product will not have any sound or vibration
    - Possible maintenance might be needed to check internal circuits due to high heat environment
    - Therefore, this constraint does not apply.

8. **Environmental constraints**
    - Places with little to no sunlight might cause problem when it comes to recharging the solar panel or battery
    - Materials such as cadmium are used in the production of solar photovoltaic cell which could be poisonous by poisoning our water if not properly disposed of
    - This constraint applies to our product because of the geographical placement of our product.

9. **Quality constraints**
    - Quality insurance that the product will be able to perform under high heat conditions
    - Testing of LED panel, radio transmission, and sensors before operable conditions to be met
    - This constraint does not apply to our project because of a long life cycle and easily replaceable parts.

10. **Ethical constraints**
    - Intellectual property using designed for purpose microchips
    - Using unrecyclable electronics would have an environmental effect at the end of lifetime of product
    - This constraint does not apply because the Engineers standards of conduct are already incorporated into the overall design process.

Section 7: Product Requirement and Specification Document

1. Introduction

This document contains the system requirements for ***SNG***. These requirements have been derived from several sources, including **Director Municipal Solid-State Street Lighting Consortium Seattle City Light | City of Seattle, Check the Air Quality in Orosi, California." BreezoMeter**.

1.1 Purpose of This Document

This document is intended to guide development of ***SNG***. It will go through several stages during the course of the project:

1. **Draft:** The first version, or draft version, is compiled after requirements have been discovered, recorded, classified, and prioritized.
2. **Proposed:** The draft document is then proposed as a potential requirements specification for the project. The proposed document should be reviewed by several parties, who may comment on any requirements and any priorities, either to agree, to disagree, or to identify missing requirements. Readers include end-users, developers, university faculty, course instructor, and any other stakeholders.
3. **Validated:** Once the various stakeholders have agreed to the requirements in the document, it is considered validated.
4. **Approved:** The validated document is accepted as an appropriate statement of requirements for the project. The developers then use the requirements document as a guide to implementation and to check the progress of the project as it develops.

1.3 Scope of the Product

This product is designed to detect and alert users of hazardous gas leaks. It will be self-powered through a solar panel and also contain an LED with a motion sensor for passing through traffic.

1.4 Case for the Product (Need)

This product is required for high traffic areas, parking lots, campus environments, and job sites. It will provide a useful service of detecting harmful gases and alerting a user immediately. It also provides energy sustainability for light posts using solar panels to power a LED at night.

2. General Description

Our team's first concept idea is a multi- gas sensor powered by a solar panel with LED for street light replacement. The device will use a solar panel array to charge a LI-ion battery that will power a Raspberry Pi microcontroller. The raspberry pi will take incoming data from multiple sensors including: gas sensors, ambient light sensors, temperature sensors, battery power sensors. The raspberry pi will organize this data and do the following: the ambient light sensor will prompt the LED to turn on in low-light conditions, the temperature sensor will give the MCU feedback on operating conditions and possible low-power set up for high heat protection, the battery power sensor will control operation times of MCU, through continuous use or time operated use, the multiple gas sensors will take in gas data of common city pollutants. The MCU will then transmit the data through a cellular transceiver to a user that can visualize the data for understanding of the consumer. The MCU can also send an alert to users if levels exceed normal conditions. Through Streamr, live streaming data network, and Smart Citizen, a platform that connects people and cities through localized data sharing, the concept of a future smart city development with use of individual homeowner air quality data that can be uploaded to a cloud server that impart when used will give back DATAcoin token which is based on Etherum block-chain to homeowner for use of their data.

2.1 Product Perspective

This product is tailored to job site managers, city officials, Department of Transportation, Environmental Protection Agency. Having an alert when there is a gas leak or fire at a job site or an overabundance of harmful car emissions on the highway can be very useful to users. This data can also be used by data scientists for prediction and logging of environmental conditions in select areas and useful for future smart city projects.

2.2 Product Functions

The main function of the SNG is to analyze environmental sensor data, log the data, and provide alerts if levels exceed normal health standards. The secondary function will be providing a standard light post powered by a solar panel.

2.3 User Characteristics

This product is intended for common person use who has a cell phone. An alert can be sent to a cell phone as a text providing details of alert and location. The technical parts of this system will be intended for scientists as it will provide environmental readings to be interpreted and analyzed through a computer.

2.4 General Constraints

The platform used was Windows 10, using various software including: Raspberry Pi IDE, Pspice Cadence PCB editor, Pspice Cadence Capture, Firefox internet browser for research and component purchase. Using a digital multimeter for circuit testing. Using a Raspberry Pi imager for operating system of the Raspberry Pi

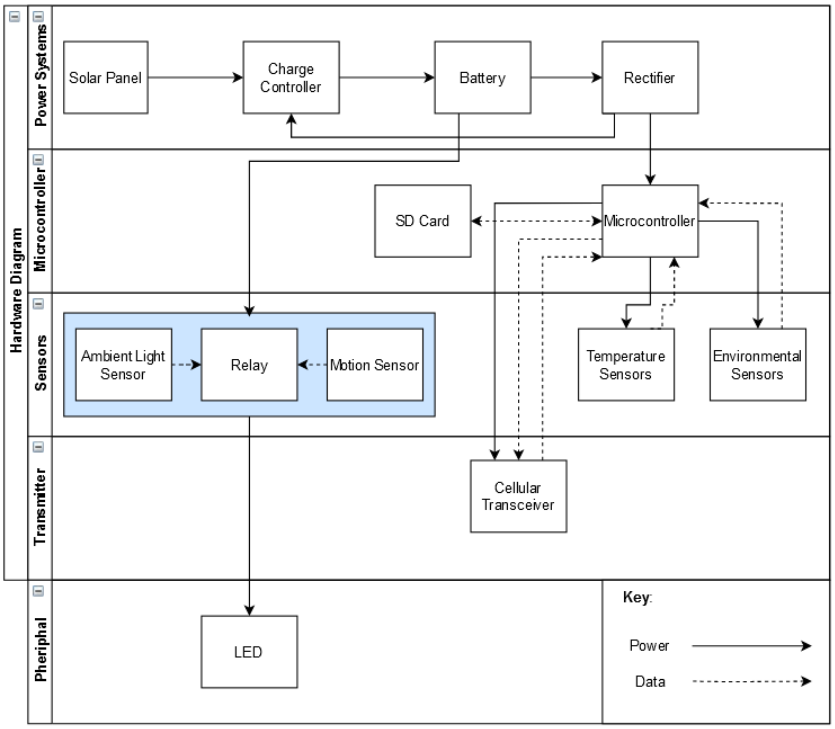
2.5 Assumptions and Dependencies

The design of our product utilizes cellular 3G/4G networks, it will be necessary that the product will be in range of a cellular tower for appropriate response of alerts or notifications. It will also have to be placed in an area with sufficient sun exposure to charge the battery. It should be placed at a correct height so it does not react to vehicle emissions.

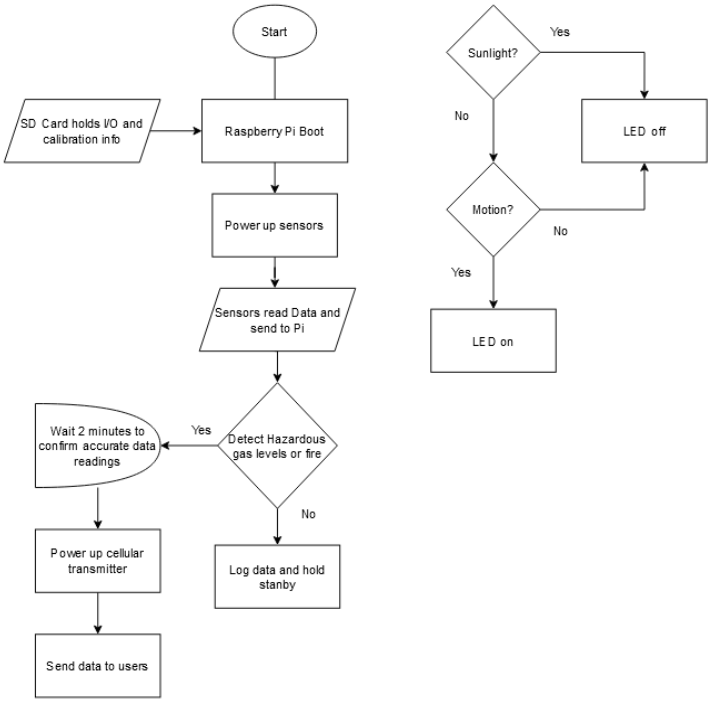
2.6 Objectives

1. Power:
    a. The device will be powered via a solar panel battery charging system. Solar panel will provide DC power that will go through a charge controller that provides overcharge protection and makes sure the solar panel does not pull power at night.
    b. The charge controller will provide a steady voltage to a battery to charge. This battery will be connected to the MCU, which will regulate through software coding.
    c. A DC/AC power inverter will provide power to the MCU and the LED fixture.
    d. The MCU will provide power to the sensors and the cellular transceiver.
2. MCU:
    a. When the MCU boots it will run the command file for our device hardware. It will prompt power to the sensors and read data.
    b. The MCU will read and log the data in an excel or another database file. It will compare the readings to a calibrated file to see if there is an excess of gas parts per million in the area.
    c. The MCU if reading an excess amount will run a command file prompting power to the cellular transceiver, the command file will also contain code initiating a text “Alert in device area” “GPS coordinates ‘ ‘ “.
    d. The MCU will then log data and alert notifications in the database file, to be pulled in the future if necessary.
3. LED:
    a. The Solar Panel battery system will also provide power to LED at night, the power coming from the battery and DC/AC inverter will go to a relay with a motion and ambient light sensor.
    b. The Solar Panel will turn on at night if there is movement in the localized vicinity.

Functional Diagram



Software Diagram



2.7 Plan

- High Level Tasks

    - Develop Solar Power management system
    - Print PCB modules for sensor array
    - Create code for sensor array retrieval and cellular transceiver
- Principles or theories to be used
    - Electrical Engineering principles on circuit design
    - C++ or python code libraries and implementation
- Significant variables/potential pitfalls/complications to be addressed
    - Battery Life
    - Distance to cell tower and user
- Design approach and methodology
    - Use known standards for Solar Panel battery charging
    - Use Microcontroller spec sheets and code
    - Test code without battery, then apply battery power for continuous use
- Integration and implementation approach
    - Parking lot light post design implementation
- Prototype validation and verification testing approach
    - Test Solar power to battery
    - Test sensor working and data retrieval
    - Test cellular response

3. Specific Requirements and Specifications

This section of the document lists specific requirements and specifications for ***SNG***. Requirements and specifications are divided into the following sections:

1. User requirements and specifications. These are requirements and specifications written from the point of view of end users, usually expressed in narrative form.
2. System requirements and specifications. These are detailed requirements and specifications describing the functions the system must be capable of doing.
3. Interface requirements and specifications. These are requirements and specifications about the user interface, which may be expressed as a list, as a narrative, or as images of screen mock-ups.

3.1 User Requirements and Specifications

3.1.1 User interface

- Can interface via wifi to Raspberry Pi for diagnosis. Customer interface through cell phone or computer text

3.1.2 Ergonomics

- Devices will be installed on light posts and will be stationary no ergonomic principles will be necessary for use.

3.1.3 Training or skills required

- No training will be needed, the device will send alerts to mobile phones or voip devices.

3.2 System Requirements and Specifications

3.2.1 Physical Characteristics

- Rectangular shape

3.2.2 Material Requirements
- Metal/plastic enclosure for entire unit

3.2.3 Electrical Requirements
- Lithium batteries, 3.2V, solar, raspberry pi

3.2.4 Abilities
- Detects gas through solar power

3.2.5 Limitations
- Capital requirements, raw material shortage, distribution issues, design patents labor costs, geographical locations too far north.

3.2.6 Equipment or materials required to use the product
- Putty, microcontrollers, remote control, 3.2V batteries, solar panels, motion/gas sensors, integrated circuits, and lithium batteries.

3.2.7 Equipment interface requirements
- Putty, motion sensors, raspberry pi

3.2.8 Handling and storage requirements
- Store at room temp, do not store below 32 degrees F and above 85 degrees C.

3.2.9 Cleaning and Sterilization

damp cloth, clean panels monthly.

3.2.10 Product maintenance and serviceability
- Remove battery corrosion, battery removal every 5-7 yrs due to voltage droppage, inspect every two months for rust, dust, foreign debris.

3.2.11 Operating parameters
- Solar power, wireless (bluetooth), battery source, raspberry pi, and ground wire.

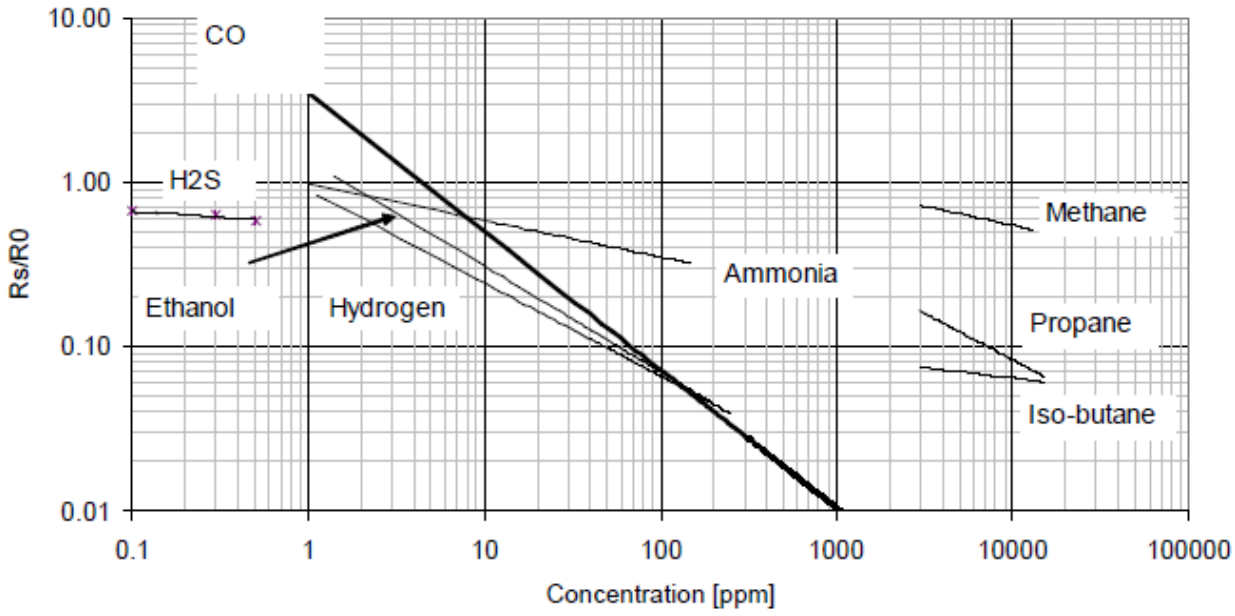
3.2.12 Repeatability and reproducibility
- Solar power is always re-usable, assuming the right geographical location and battery consumption has no issues.

3.2.13 Reliability
- Solar energy doesn't require a lot of energy and is cost-effective.

3.2.14 Mechanical safety features
- Low voltage, IC chips will not short circuit, solar panels are weather-proof.

3.2.15 Electrical safety features
- The stationary fixture will be grounded, low voltage; electric shock warning label.

3.3 Interface Requirements and Specifications



3.4 Environmental Conditions

3.4.1 Temperature: -40C to 85C
3.4.1.1 Operating: -40C to 85C
3.4.1.2 Storage: rectangular plastic enclosure
3.4.2 Humidity: 0-100%
3.4.2.1 Operating: -40C to 85C
3.4.2.2 Storage: rectangular plastic enclosure.
3.4.3 Shipping, transportation vibration

- Cardboard box, commercial vehicle/airlines

3.4.4 Pressure and Altitude

- Pressure of 9000mb, 500m agl

3.4.5 Electromagnetic Interference

- Emi is susceptible

3.4.6 Electrostatic Discharge

- Susceptible to esd

3.4.7 Impact Resistance

- Connect a FSR (force sensitive resistor)

3.5 Manufacturing

3.5.1 Cost

- Cost is unfavorable, due to supply and demand.
- Components:
- Solar panel 12.99

- Motion sensor 11.65
- Charge controller 13.49
- Liion battery 64.49
- Dc AC converter 23.89
- COmponent fan 24.99
- MCU Raspberry pi 3b+
- CO2 Ch4 C2H5OH sensor 42.99
- NO H2 sensor 24.82
- C3H8 NH3 sensor 28.95
- Cellular transceiver 36.24
- Total 326.5

3.5.2 Environmental requirements for production
- Must be in the lower 48, where sunlight is prevalent.

3.5.3 Raw materials and suppliers
- Metal, plastic, solar panels, batteries, microcontroller, silicon(high demand), copper.

3.5.4 Test methods, standard
- Hantek, breadboard

3.6 Packaging

3.6.1 Packaging configurations
- 26”x16.5”x8.4”, 17-20 lbs., rectangular panel shape.

3.6.2 Packaging materials
- Cardboard box, remote control, instruction manual.

3.6.3 Special shipment requirements

3.7 Labeling

3.7.1 Detail intended use, warning, directions for use, cleaning, expiration date
- Intended use is to receive hazardous warning of hazardous oxygen intake and gas, via solar powered panels and sensors.
- Electrical shock warning. Panels must be oriented in the direction where the most sun can be received (South), in order to charge before intended use; make sure to ground the ground wire; let charge for a few hours, then turn on the on/off switch. clean with damp cloth and remove dirt/debris every two months. also, when replacing batteries, use the proper batteries and make sure the positive/negative terminals are aligned properly; never use an old battery with a new one. The expiration date is every 10 years.

3.7.2 Identify target audience for labeling
- Civilians with lung deficiencies, smokers, covid patients, supporters of climate change, and the environmental protection agency (epa).

3.7.3 Language requirements
- ASCII, Java, C++, python

3.8 Regulatory

3.8.1 Clinical trials

- Some parts are made in China, so warranty is most likely not valid; product is very bright, cost efficient, and highly recommended; installation is difficult for one person, so two people recommended.

3.8.2 Submission type

N/A

3.8.3 CE mark

N/A

3.8.4 US and international standards

- IEEE standards, US metric standard

3.8.5 Patent issues

- No patent issues; consumers should check all patent licenses before a product is built, manufactured, and distributed.

3.8.6 Existing technology

- To avoid outdoor light with positionable solar collector, solar energy-operated street lamp system, solar powered light pole and led light fixture, and energy efficient solar powered outdoor lighting fixture system; all have official patents.

Section 8: Engineering codes and Standards

The IEEE Code of Ethics contributes an integral and vital part in the development, distribution, and maintenance of current and new technology. As well, it serves as a pivotal protection to potential consumers and those who may be impacted by the most recent technological developments. A team member's adherence to this code represents an unspoken agreement between consumers and developers regarding a safe and ethical practice when using and creating a device or program with the hopes to not only promote newer innovations but to also showcase the honest means of data and use of the product.

Our team's software-centric side understands that following the code's subsections 1, 2, 5, and 6 provide the developer-to-user protection regarding not only what the product will be capable of in its design regarding its technical components, but also what user information will be stored and used. These sections are embedded safeguards to prevent abuse and breach of information amongst other forms of malicious intent with the product by the design team. However, subsection 5 in particular provides the ongoing maintenance and correction of errors that are found with the product; things that can be found such as backdoor access and unsecured connections that can be hacked with software and network systems. To our electrical-centric side of our team we see the use of subsection 5 as guideline to maintain the hardware design and fix all manufacturing defects that are either inherent to the design or mistakes made with the individual production.

Subsections 3 and 4 prevent conflicts of interest from being intentionally or inadvertently violated. Criminal, negligent, or unprofessional behavior or actions that are reflected in the

product or treatment of others are unacceptable. Prevention of incidents such as this will allow the team, the company, consumer, or any future investors from being held accountable for untoward or illegal actions by parties who engage in such activity from being associated with the parties listed above. As well, providing our list of associates to the public will keep our team and company above board and clearly communicate the intentions of such groups. Subsections 7, 8, and 9 are integral to the internal relationships between developers as employees working as a team. Discrimination in the workplace between employees represent an unhealthy and unethical environment acting as an extreme detriment to the overall welcoming and inclusive atmosphere that should be fostered in a STEM field. These subsections, as well as all others, are upheld by subsection 10 to protect employees who do report violations of the code. This will ensure accountability between all parties in the event a violation occurs.

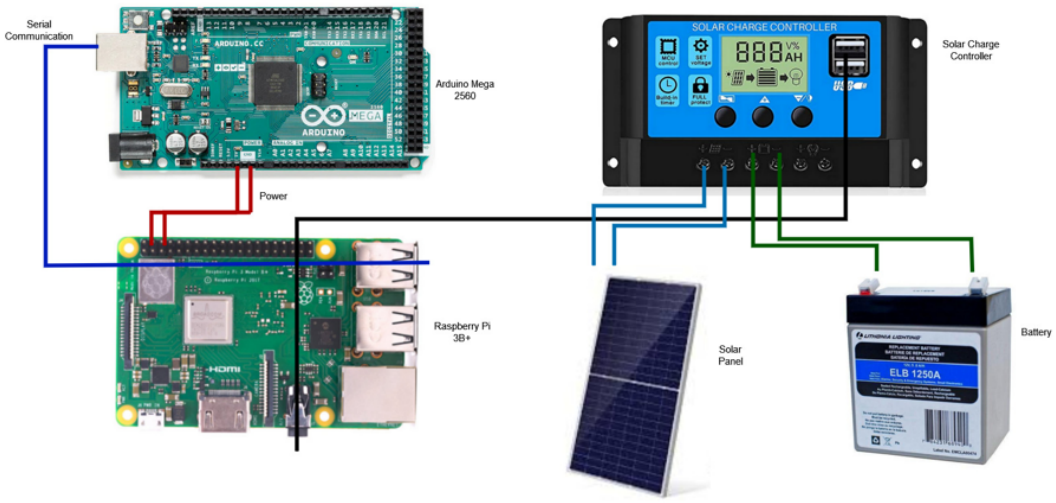
Section 9: Design Detail

The SNG was built with several hardware components, we had multiple sensors and an Arduino mega, Raspberry pi 3B+. The sensor was connected to the Arduino Mega. We also had the Node Esp8266 to connect the device to the internet.

We share responsibility for the components and tasks based on the skill level and expertise of the members. We acquired Led and solar panels and decided on the charge controller, we then acquired a Gratury junction box as the housing unit which is a temperature, oil, chemical and water resistant box.

Plan

- Develop gas sensing solar panel light pole mount
    - Integrate multiple languages and IDEs
    - Assemble circuits to connect all components
    - Test sensors for sensitivity
    - Test solar panel for charge time and capacity
- Develop software for communicating sensor output to terminal for live data acquisition
    - Adjust sensor offsets
    - Determine means of communication (serial, wired or wireless connection)
- Assemble mount
    - Modify housing unit with holes for cooling and pulling in particulates
    - Fit components to inner walls for stable transport
    - Manage wires

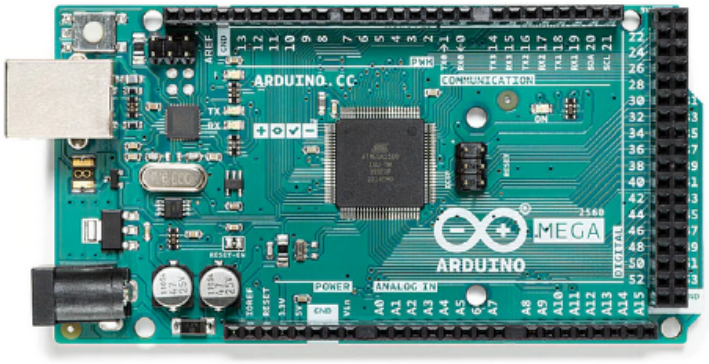


Section 10: Major Component

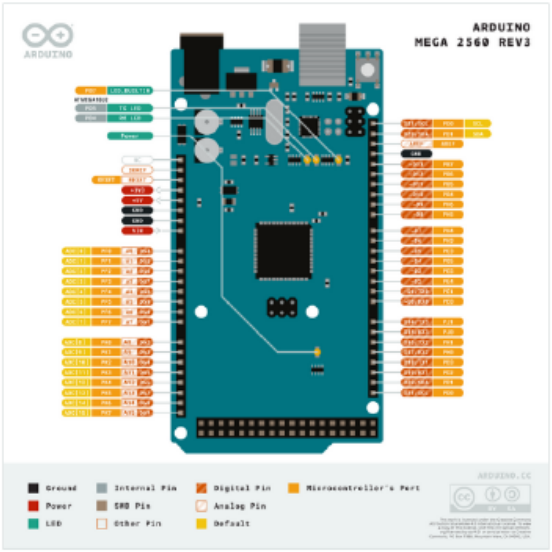
Arduino Mega 2560 Rev3

The Arduino mega 2560 is one of the microcontrollers we decide to use for our project. First, the Arduino 2560 54 digital I/O's, (15 used as PWM outputs), 16 analog inputs, 4 UARTS, 16 MHz crystal oscillator USB port, powerjack, ICSP header, and a reset button. For the clock speed, the main processor is 16 MHz, the CPU is an Atmel AVR (8 bit), ARM Cortex-MOT(32 bit), and the memory is SRAM. Arduino's are programmed using C and C++ And is used with sensors, actuators, thermostats, and motion detectors.

Arduino Mega 2560 Rev3



Pinout Diagram



Raspberry pi 3B+

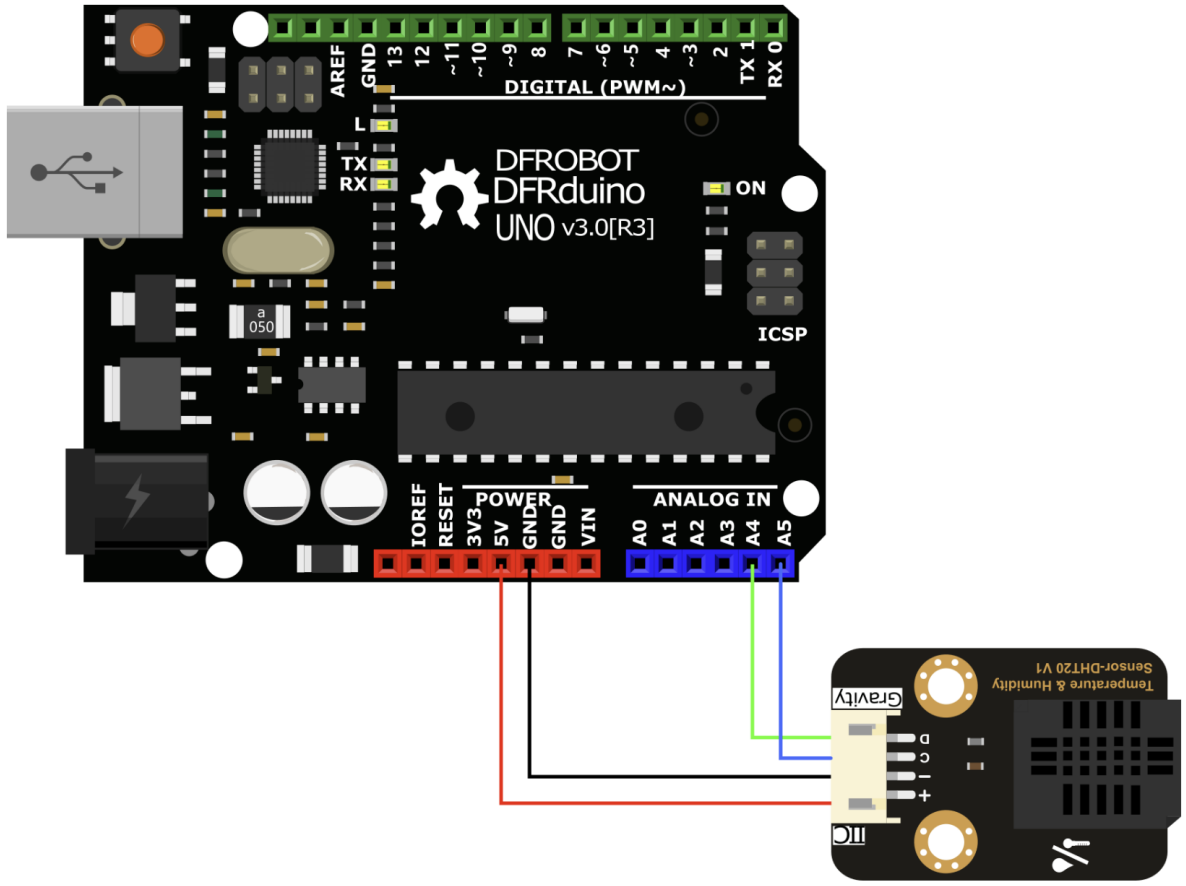
The Raspberry pi 3B+ has a 1.4 GHz processor, 3x faster gigabit ethernet, USB 2.0 @ 300 Mb/s, 2.4 GHz dual-band 802.11 ac, wifi (100Mb/s), POE hat, A Broadcom BCM 2837 SOC, 64 bit quad-core ARM cortex=A53 processor, with a 312 kb shared L2 cache. It can be overclocked at 800 MHz. The overclocking modes are: 1100 MHz ARM, 550 MHz core, 500 MHZ SDRAM, and 6 overvolting. This microcontroller mainly uses python computer coding/language.

DHT20 Temperature/Humidity Sensor

The DHT20 Sensor is connected via pins: A4 from Arduino 2560 to SDA(data line) on the DHT20 Sensor, 5V pin on the Arduino 2560 to the positive pin on the DHT20, GND on the Arduino 2560 to the negative pin on the DHT20 Sensor, and A5 on the Arduino 2560 to SCL(clock) on the DHT20 Sensor in a parallel circuit diagram.

Mult

Connection Diagram



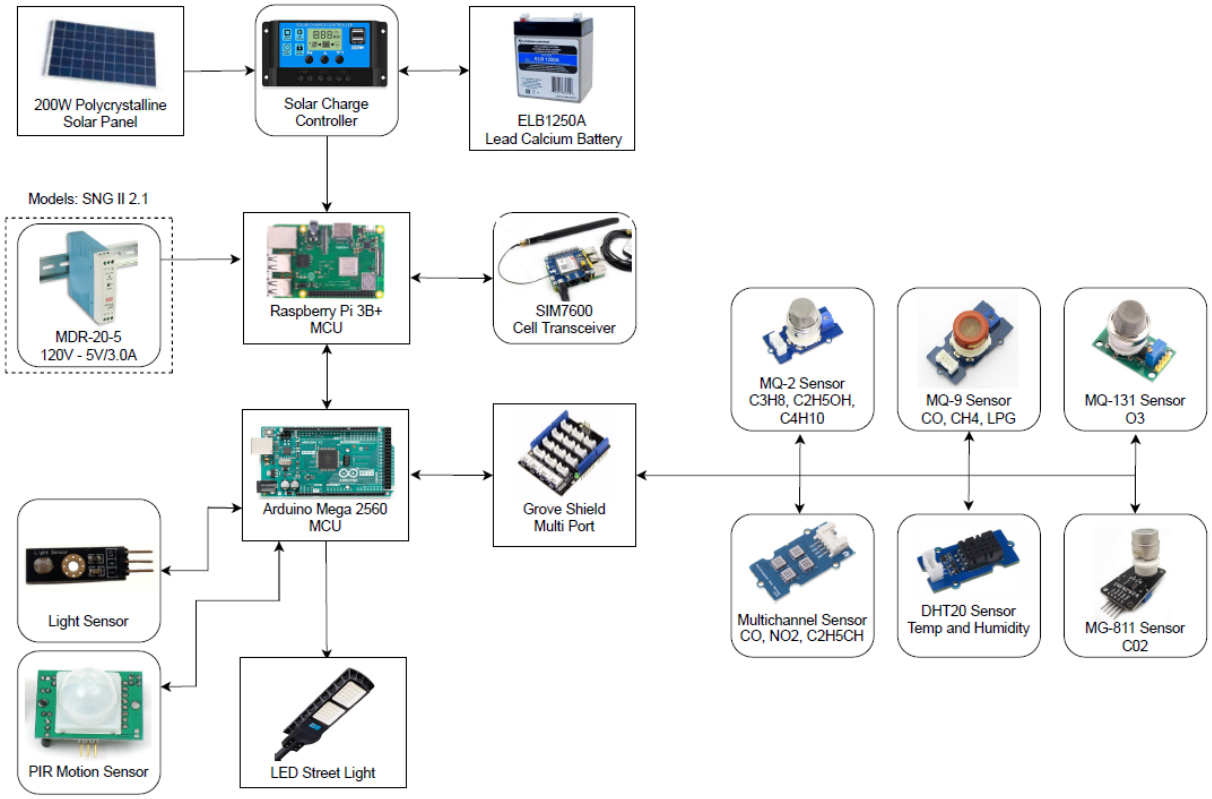
Grove Multichannel sensor

The multichannel sensor provides stable and reliable gas detection for 4 different kinds of gases. It can detect Carbon monoxide (CO), Nitrogen dioxide (NO2), Ethyl alcohol(C2H5CH), Volatile Organic Compounds (VOC), and etc. It is connected to the Arduino mega board. This

sensor evaluates GM-102B, GM-302B, GM-502B, and GM-702B sensors. The gas sensor is based on MEMS technology and is more suitable for qualitative than quantitative measurement.

Grove Gas Sensor MQ9,MQ2

Section 11: Hardware Block Diagram

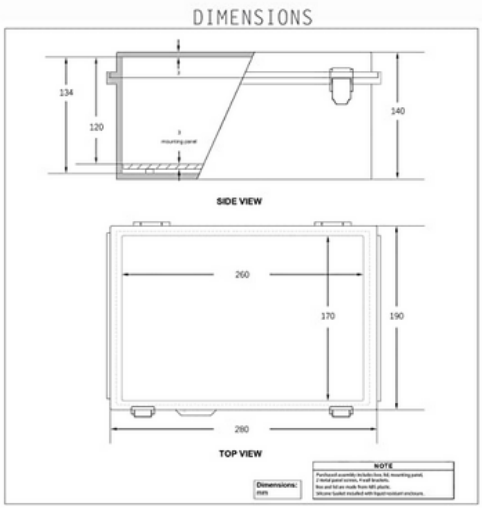


Section 12: Enclosure Design

For the enclosure box for the sensors, microcontroller, battery, and charge controller we used a Gratury junction box, hinged cover stainless latch waterproof plastic enclosure that included a mounting plate and wall bracket (11.4x7.5x5.5). It was finished with a

gray exterior. It is temperature resistant so whenever it gets hot it doesn't affect the sensors or microcontrollers inside. It is oil and chemical resistant.

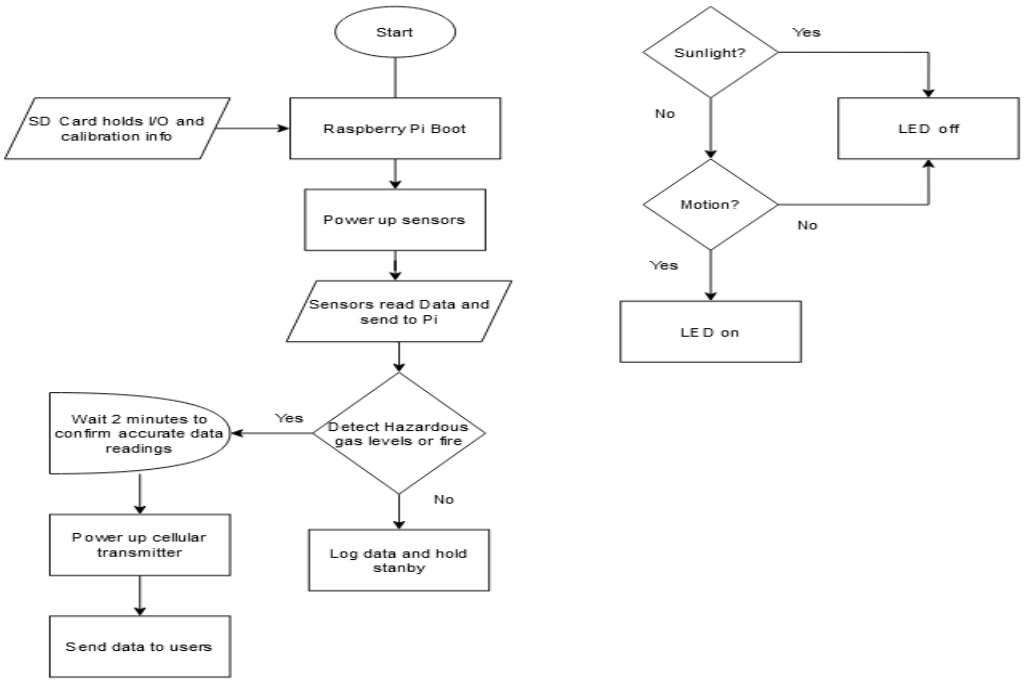








Sensor 13: **<u>Software Block Diagram</u>**



Sensors Software

Arduino IDE



```
#include "Air_Quality_Sensor.h"                                  //  AirQuality
AirQualitySensor sensor(A0);                                     //  AirQuality

#include <Multichannel_Gas_GMXXX.h>                              //  Multichannel
#ifdef SOFTWAREWIRE                                              //  Multichannel
        #include <SoftwareWire.h>                                //  Multichannel
        SoftwareWire myWire(3, 2);                               //  Multichannel
        GAS_GMXXX<SoftwareWire> gas;                             //  Multichannel
#else                                                            //  Multichannel
        #include <Wire.h>                                        //  Multichannel
        GAS_GMXXX<TwoWire> gas;                                  //  Multichannel
#endif                                                           //  Multichannel

#include "Wire.h"                                                //  Temp and
Humidity
#include "DHT.h"                                                         //  Temp and
Humidity
#define DHTTYPE DHT20                                            //  Temp and
Humidity

DHT dht(DHTTYPE);                                                //  Temp and
Humidity

#if defined(ARDUINO_ARCH_AVR)                                    //  Temp and
Humidity
        #define debug  Serial                                    //  Temp and
Humidity

#elif defined(ARDUINO_ARCH_SAMD) ||  defined(ARDUINO_ARCH_SAM)//   Temp and
Humidity
        #define debug  SerialUSB                                 //  Temp and
Humidity
#else                                                            //  Temp and
Humidity
        #define debug  Serial                                    //  Temp and
Humidity
#endif                                                           //  Temp and
Humidity

void setup() {
```

```
	debug.begin(9600);                                                  //  Temp and
Humidity
	Wire.begin();                                                       //  Temp and
Humidity
	dht.begin();                                                        //  Temp and
Humidity

	Serial.begin(9600);                                                 //  Multichannel
	gas.begin(Wire, 0x08);                                              //  Multichannel

	Serial.println("CLEARDATA");                    //  Excel
	Serial.println("LABEL,Time, Humidity %, Temperature *F, NO2 ppm, C2H5OH ppm, VOC
ppm, CO ppm, MQ9 RS_ratio, MQ9 Rs/R0");                             //  Excel

}

void loop() {


	int quality = sensor.slope();                                       //  AirQuality

	if (quality == AirQualitySensor::FORCE_SIGNAL) {                            //  AirQuality

	} else if (quality == AirQualitySensor::HIGH_POLLUTION) {           //  AirQuality

	} else if (quality == AirQualitySensor::LOW_POLLUTION) {            //  AirQuality

	} else if (quality == AirQualitySensor::FRESH_AIR) {                //  AirQuality

	}                                                                   //  AirQuality

	delay(100);                                                         //  AirQuality

	int row_excel = 0;                                                  //  Excel
	row_excel++;                                                        //  Excel
	Serial.print("DATA,TIME,");                                                 //  Excel


	float temp_hum_val[2] = {0};                                        //  Temp and
Humidity
	if (!dht.readTempAndHumidity(temp_hum_val)) {                       //  Temp and
Humidity
	Serial.print(temp_hum_val[0]);                                      //  Temp and
Humidity
```

```
        Serial.print(",");                                      //  Temp and
Humidity
        temp_hum_val[1]=temp_hum_val[1] * 1.8 + 32;             //  Temp and
Humidity
        Serial.print(temp_hum_val[1]);                          //  Temp and
Humidity
        Serial.print(",");                                      //  Temp and
Humidity
        }                                                       //  Temp and
Humidity
        delay(1500);                                            //  Temp and
Humidity

        uint8_t len = 0;                                        //  Multichannel
        uint8_t addr = 0;                                       //  Multichannel
        uint8_t i;                                              //  Multichannel
        uint32_t val = 0;                                       //  Multichannel

        val = gas.measure_NO2();                                //  Multichannel
        Serial.print((val));                                    //  Multichannel
        Serial.print(",");                                      //  Multichannel
        val = gas.measure_C2H5OH();                             //  Multichannel
        Serial.print((val));                                    //  Multichannel
        Serial.print(",");                                      //  Multichannel
        val = gas.measure_VOC();                                //  Multichannel
        Serial.print((val));                                    //  Multichannel
        Serial.print(",");                                      //  Multichannel
        val = gas.measure_CO();                                 //  Multichannel
        Serial.print((val));                                    //  Multichannel
        Serial.print(",");                                      //  Multichannel
        delay(1500);                                            //  Multichannel


        float sensor_volt2;                                     //  MQ9
        float RS_gas2; // Get value of RS in a GAS              //  MQ9
        float ratio2; // Get ratio RS_GAS/RS_air                //  MQ9
        int sensorValue2 = analogRead(A1);                      //  MQ9
        sensor_volt2=(float)sensorValue2/1024*5.0;              //  MQ9
        RS_gas2 = (5.0-sensor_volt2)/sensor_volt2;              //  MQ9
        ratio2 = RS_gas2/0.09;  // ratio = RS/R0                //  MQ9
        Serial.print((RS_gas2));                                //  MQ9
        Serial.print(",");                                      //  MQ9
        Serial.print((ratio2));                                 //  MQ9
        Serial.print(",");                                      //  MQ9
```

```
        delay(1500);                                                    //  MQ9

float sensor_volt;                                                      //  MQ2
        float RS_gas; // Get value of RS in a GAS                       //  MQ2
        float ratio; // Get ratio RS_GAS/RS_air                         //  MQ2
        int sensorValue = analogRead(A2);                               //  MQ2
        sensor_volt=(float)sensorValue/1024*5.0;                        //  MQ2
        RS_gas = (5.0-sensor_volt2)/sensor_volt;                        //  MQ2
        ratio = RS_gas/0.09;  // ratio = RS/R0                         //  MQ2
        Serial.print((RS_gas));                                         //  MQ2
        Serial.print(",");                                              //  MQ2
        Serial.print((ratio2));                                         //  MQ2
        Serial.print(",");                                              //  MQ2
        delay(1500);                                                    //  MQ2


        Serial.println(row_excel);                                      //  Excel


}
```

Section 14: Major Problem and Conclusion

The sensor had to be on for a certain amount of time before starting to acquire accurate data, we had issues with power distribution to different microcontrollers and sensors since they all required a different voltage. Overall the initial project's scope was a functional solar panel LED lamp post but through challenging additions and modifications, the integration of the solar panel with its gas sensors came to its fruition. Its current stage of design and production leave a decent amount of improvement and refinement to be made as well as ample room for innovations which was always the end goal for its design. Expansive additions, wide use application, and easy modification were the core tenets of the project.